

Part: **4.4 – CRACK PROPAGATION ANALYSIS**

Chapter: **4410**

Title: **STRESS INTENSITY FACTOR (K)**

Revision: **0.1**

Date: **24/09/2021**

1 INTRODUCTION

The stress intensity factor, abbreviated as SIF and represented by K, is the magnitude that defines the severity of the stress distribution around the crack tip. It is used to establish failure criteria due to material fracture and to calculate the crack growth rate.

It is important to highlight that K, in the field of crack propagation, is not an analogous concept of K_t in the field of crack initiation. The stress concentration factor is a dimensionless factor which accounts only for the geometry and indicates how the stress distribution changes around a notch, while the stress intensity factor has units (ie: $MPa\sqrt{mm}$) and represents the severity of the stress distribution around the crack tip.

The stress field for a crack in an infinite sheet loaded in tension can be obtained in polar coordinates using the conventional theory of elasticity as:

$$\sigma_x = \frac{\sigma \cdot \sqrt{\pi \cdot a}}{\sqrt{2 \cdot \pi \cdot r}} \cdot \cos \frac{\theta}{2} \left(1 - \sin \frac{\theta}{2} \cdot \sin \frac{3\theta}{2} \right)$$

$$\sigma_y = \frac{\sigma \cdot \sqrt{\pi \cdot a}}{\sqrt{2 \cdot \pi \cdot r}} \cdot \cos \frac{\theta}{2} \left(1 + \sin \frac{\theta}{2} \cdot \sin \frac{3\theta}{2} \right)$$

$$\tau_{xy} = \frac{\sigma \cdot \sqrt{\pi \cdot a}}{\sqrt{2 \cdot \pi \cdot r}} \cdot \cos \frac{\theta}{2} \cdot \sin \frac{\theta}{2} \cdot \sin \frac{3\theta}{2}$$

These equations are accurate for small values of r. This means that they are a good approach of the stress field around the crack tip (see Figure 1).

The above equations can be written as:

$$\sigma_{i,j} = \frac{K}{\sqrt{2 \cdot \pi \cdot r}} \cdot f_{i,j}(\theta)$$

Where:

$$K = \sigma \cdot \sqrt{\pi \cdot a}$$

This equation of the SIF is valid for an infinite geometry with a central crack of length $2a$ under tension load. To account for other geometries and loading conditions it is necessary to introduce a dimensionless correction factor: the so-called normalized stress intensity factor β :

$$K = \beta \cdot \sigma \cdot \sqrt{\pi \cdot a}$$

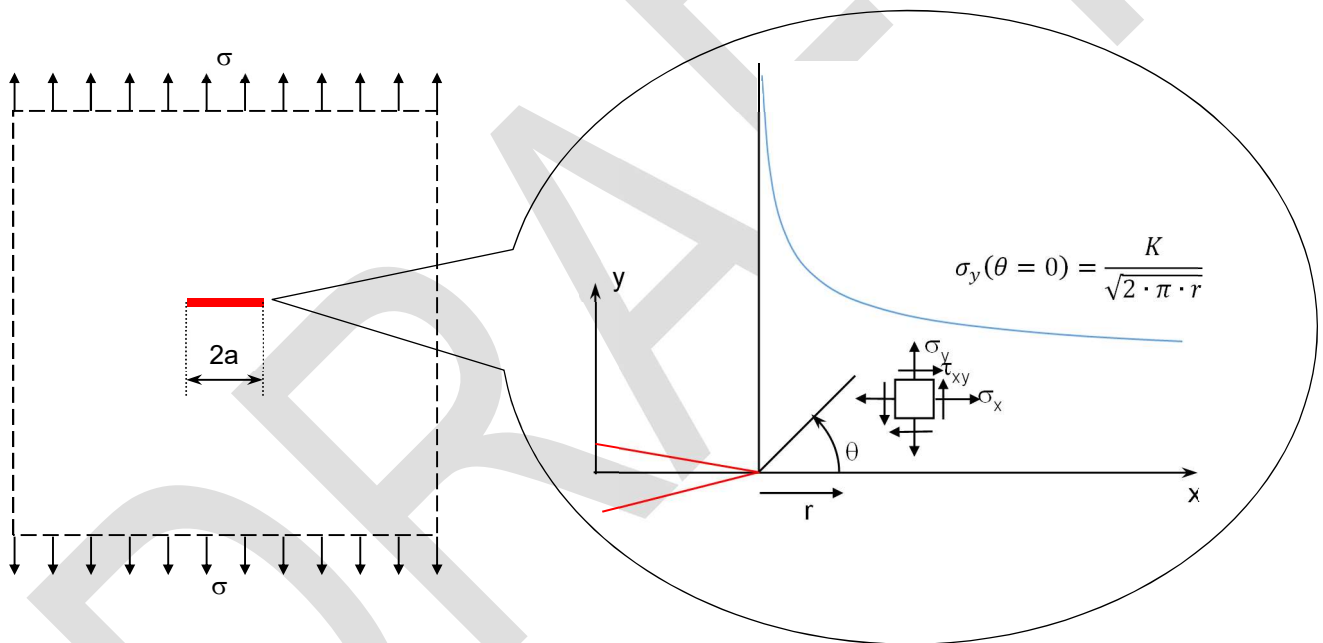


Figure 1. Stress field around the crack tip

2 SIF CALCULATION PROCEDURE

The stress intensity factor solutions for typical geometries and crack configurations can be found in open literature such as Ref. 1, Ref. 2 and Ref. 3.

These solutions can be used to obtain the solution of more complex configurations by using the superposition principle or compounding method.

SUPERPOSITION PRINCIPLE

For complex load conditions, the stress intensity factor can be determined by linearly adding the stress intensity factor due to each type of loading applied to the same geometry. It is important to highlight that the process of adding SIF solutions can only be applied to SIF associated to the same structural and crack geometry.

$$K = K_1 + K_2 + \dots + K_n$$

$$K_1 = \beta_1 \cdot \sigma_1 \cdot \sqrt{\pi \cdot a}$$

$$K_2 = \beta_2 \cdot \sigma_2 \cdot \sqrt{\pi \cdot a}$$

$$K_n = \beta_n \cdot \sigma_n \cdot \sqrt{\pi \cdot a}$$

The resulting β factor would be defined in the following way:

$$\beta = \beta_1 \cdot \frac{\sigma_1}{\sigma} + \beta_2 \cdot \frac{\sigma_2}{\sigma} + \dots + \beta_n \cdot \frac{\sigma_n}{\sigma}$$

An example is shown in Figure 2. The proposed configuration (K) has not an available analytical solution. To calculate the SIF the problem has to be decomposed in the cases 1 (K_1) and 2 (K_2). The SIF for the case 1 (K_1) is a well-known solution, but the solution for K_2 has to be estimated by summing and splitting load cases (see chapter 5.4 of Ref. 5):

$$K = K_1 + K_2 = K_1 + \frac{K_3 + K_4}{2}$$

$$K_1 = \beta_1 \cdot \sigma_2 \cdot \sqrt{\pi \cdot a}$$

$$K_3 = \beta_1 \cdot (\sigma_1 - \sigma_2) \cdot \sqrt{\pi \cdot a}$$

$$K_4 = \beta_2 \cdot \frac{P}{2a} \cdot \sqrt{\pi \cdot a}$$

$$K = \sigma_1 \cdot \sqrt{\pi \cdot a} \cdot \left(\beta_1 \cdot \frac{\sigma_2}{\sigma_1} + \frac{1}{2} \cdot \left(\beta_1 \cdot \frac{(\sigma_1 - \sigma_2)}{\sigma_1} + \beta_2 \cdot \frac{P/2a}{\sigma_1} \right) \right)$$

Where β_1 is the solution for a centred crack in a finite width sheet under tension load and β_2 is the solution for a crack in a finite width sheet with two loads on the crack edges. Both solutions can be found in open literature.

STRESS INTENSITY FACTOR (K)

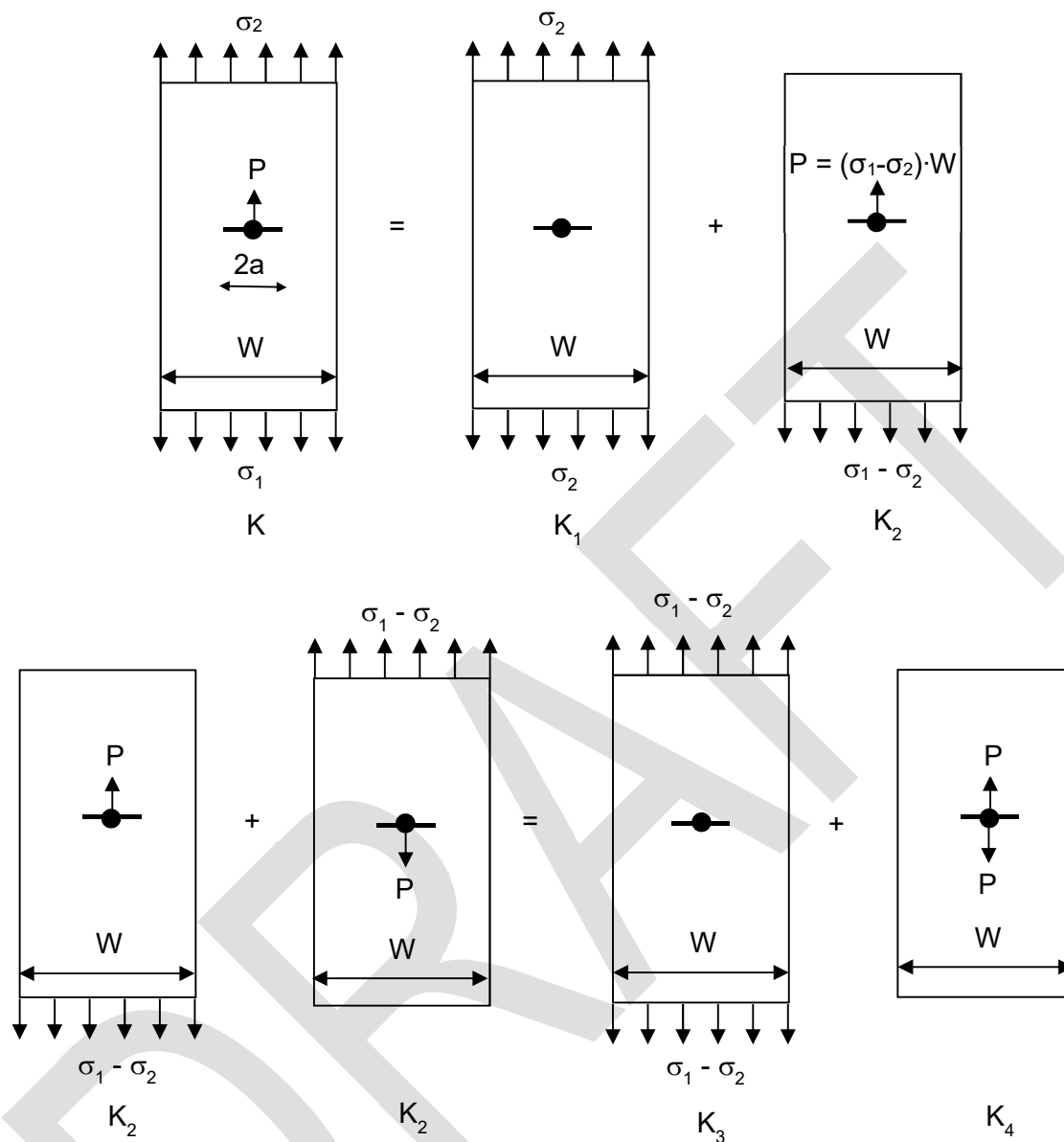


Figure 2. Stress Intensity Factor for a cracked sheet ($t = 1$) with pin-loaded hole and tension stress

COMPOUNDING METHOD

For complex geometries, the problem has to be splitted into simpler configurations, each of these containing only one geometric effect that interacts with the crack and having known solutions. The final normalized SIF (β) is then calculated as the product of all of them:

$$\beta = \beta_1 \cdot \beta_2 \cdot \beta_3 \cdots \beta_n$$

For example; consider an edge crack in a finite width sheet under tension load, growing towards a hole:

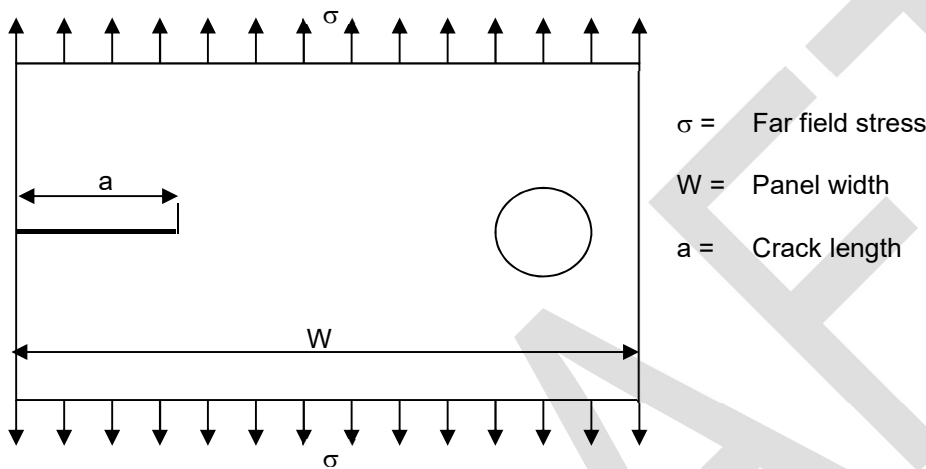


Figure 3. Crack in a finite width sheet growing towards a hole

The SIF for the crack tip growing towards the hole would be calculated as:

$$K = \beta_1 \cdot \beta_2 \cdot \sigma \cdot \sqrt{\pi \cdot a}$$

Where: β_1 accounts for a through crack growing from a free edge in a finite width panel with constant thickness, subjected to a constant far field stress distribution, normal to the crack at both sides of the section.

β_2 accounts for a crack growing towards a hole in an infinite width panel with constant thickness, subjected to a constant far field stress distribution, normal to the crack, at both sides of the section.

SIF SOLUTIONS FOR COMPLEX CONFIGURATIONS

There are more complex configurations, for which the SIF cannot be obtained by compounding the solution for simple geometries or load conditions. In these cases, the stress intensity factor can be calculated by using numerical analysis procedures such as:

- Finite element method
- The boundary collocation technique
- The boundary integral method
- The displacement compatibility method; this method is used to calculate the stress intensity factor of stiffened panels (Ref. 6).
- The Bückner weight function technique; the weight function technique is used to calculate the stress intensity factor of a cracked component under a non-constant stress distribution, based on the stress distribution obtained for the uncracked part.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Crack length
β	-	Normalized Stress Intensity Factor
K, SIF	MPa $\sqrt{\text{mm}}$	Stress Intensity Factor
Kt	-	Stress Concentration Factor
LEFM	-	Linear Elastic Fracture Mechanics
P	N	Load
σ	MPa	Stress
r	mm	Distance from the crack tip
W	mm	Panel width

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[P. Rooke and D. J. Cartwright]		Compendium of Stress Intensity Factors
Ref. 2	[NASGRO]	2016	Reference Manual Version 8.01
Ref. 3	H.Tada, P.Paris, G.Irwid	1985	The stress analysis of cracks handbook
Ref. 4	Feddersen C.E.	1966	Discussion in ASTM STP 410
Ref. 5	Schivje J.	2 nd Edition	Fatigue of Structures and Materials
Ref. 6	Swift T.	ASTM ST 842, 1984	Fracture Analysis of Stiffened Structure

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
0.1 24/09/2021	M.Carmen.Blanquer Darío Fernández	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEPAA)	Javier Gómez-Escalonilla Xx/xx/2021	Holger Hickethier Xx/xx/2021
Stress Methods (TEPAP-TL4)	Darío R. Fernández Villalba Xx/xx/2021	-

Table 3. *Approval signatures table for last revision of this chapter*